# Drop-in text for Sec. VI-E (deviation bound) and Sec. VI-B (ramp tracking)

numbering continues from Eq. (86); symbols follow the manuscript

## VI-E continuation: impulse response and the bound on $e_\varphi$

*After Proposition 1 / Eq. (86).*

The homogeneous part of (86) is spanned by $\cosh C_2\tau$ and $\sinh C_2\tau$ with $C_2 = \sqrt{Wz_{CoM}/J_P}$ from (20). Its impulse response — the solution of $J_P\ddot{h} - Wz_{CoM}\,h = \delta(\tau)$ from rest — follows from the momentum jump $J_P[\dot{h}(0^+) - \dot{h}(0^-)] = 1$, the position being unable to jump:

$$h_\varphi(\tau) = \frac{\sinh(C_2\tau)}{J_P C_2}, \qquad h_\varphi(0) = 0, \qquad \dot{h}_\varphi(0) = \frac{1}{J_P}. \tag{87}$$

Because $h_\varphi(0) = 0$, the boundary term generated by differentiating a convolution with a variable upper limit,

$$\frac{d}{d\tau}\int_0^\tau h_\varphi(\tau - s)\rho(s)\,ds = \underbrace{h_\varphi(0)}_{=0}\rho(\tau) + \int_0^\tau \dot{h}_\varphi(\tau - s)\rho(s)\,ds,$$

vanishes for *any* forcing. The zero initial conditions in (86) therefore set both homogeneous coefficients to zero — no assumption on $\rho(0)$ is required — and the deviation is the Duhamel integral alone,

$$e_\varphi(\tau) = \int_0^\tau h_\varphi(\tau - s)\,\rho(s)\,ds, \qquad e_\omega(\tau) = \int_0^\tau \dot{h}_\varphi(\tau - s)\,\rho(s)\,ds. \tag{88}$$

**The two channels.** Two terms of (79) leave the estimator span: the gravity remainder (83) and the bilinear ground-effect coupling (79)(iv),

$$\rho(\tau) = \rho_\varphi(\tau) + \rho_{GE}(\tau), \qquad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\varphi. \tag{89}$$

Both vanish at the onset — with $\varphi_{\mathrm{nom}} \sim \tau^3$ the gravity remainder is $\mathcal{O}(\varphi^2) \sim \tau^6$ and the bilinear term is $\tau \cdot \tau^3 = \tau^4$ — so each is largest at the window end.

**An envelope bound that can be verified by hand.** The estimator consumes the *rate*: the two-segment fit is applied to $\omega = C_1(\cosh C_2\tau - 1)$ reconstructed from the gyro, and the onset is the argument that minimises its residual. The quantity that must be bounded is therefore $e_\omega$, and $e_\varphi$ enters only through the gravity remainder that generates $\rho$. Replacing $\rho$ by a constant $\bar\rho$ and integrating the two kernels, $\int_0^\tau \dot{h}_\varphi = \sinh(C_2\tau)/(J_P C_2)$ and $\int_0^\tau h_\varphi = (\cosh C_2\tau - 1)/(J_P C_2^2)$, with $J_P C_2 = Wz_{CoM}/C_2$ and $J_P C_2^2 = Wz_{CoM}$,

$$\big|e_\omega(\tau_{\mathrm{end}})\big| \le \frac{\bar\rho\, C_2}{Wz_{CoM}}\sinh x, \qquad \big|e_\varphi(\tau_{\mathrm{end}})\big| \le \frac{\bar\rho}{Wz_{CoM}}\big(\cosh x - 1\big), \qquad x \triangleq C_2\tau_{\mathrm{end}}. \tag{90}$$

The inertia cancels in both. Only $Wz_{CoM}$, $C_2$, the dimensionless group $x$ and the forcing level enter.

**The window length is bounded a priori by the tilt cap.** Neither channel maximum nor the kernel factor requires $x$ to be solved for. With the nominal excursion of (20),

$$\varphi_{\mathrm{end}} = \frac{\dot{M}}{Wz_{CoM}C_2}\Lambda(x), \qquad \Lambda(x) \triangleq \sinh x - x = \sum_{k\ge 1}\frac{x^{2k+1}}{(2k+1)!}. \tag{91}$$

Every term of $\Lambda$ is positive, so $\Lambda(x) \ge x^3/6$, and since $\Lambda$ increases, imposing the tilt cap $\varphi_{\mathrm{end}} \le \varphi_{\max}$ inverts in closed form:

$$x \le \bar{x} \triangleq \Big(\frac{6\,\varphi_{\max}Wz_{CoM}C_2}{\dot{M}}\Big)^{1/3}, \qquad \tau_{\mathrm{end}} \le \frac{\bar{x}}{C_2} = \Big(\frac{6\,\varphi_{\max}Wz_{CoM}}{C_2^2\dot{M}}\Big)^{1/3}, \tag{92}$$

the second form using $J_P = Wz_{CoM}/C_2^2$. The retained term is the inertia-only rise $\varphi = \dot{M}\tau^3/(6J_P)$ obtained by dropping the destabilising $Wz_{CoM}\varphi$: it is the slowest admissible ascent and therefore reaches the cap latest, which is why (92) bounds $x$ from above and why the worst case is the *slowest* ramp, not the fastest. The moment increment over the window and the coupling channel follow with no fitted quantity at all,

$$\begin{aligned} \Delta M_{\text{win}} &= \dot{M}\tau_{\text{end}} \;\le\; \big(6\,\varphi_{\max} J_P\,\dot{M}^2\big)^{1/3}, \\ \rho_{GE,\max} &= |\beta_M|\,\Delta M_{\text{win}}\,\varphi_{\text{end}} \;\le\; |\beta_M|\,\varphi_{\max}\big(6\,\varphi_{\max} J_P\,\dot{M}^2\big)^{1/3}. \end{aligned} \tag{93}$$

and the kernel factors of (90) close on the same constant through the identity $\cosh x - 1 = \Lambda(x) + (x - 1 + e^{-x})$ with $0 \le x - 1 + e^{-x} < x$, together with $\sinh x = \Lambda(x) + x$:

$$\cosh x - 1 \;\le\; \sinh x \;\le\; \Lambda(x) + \bar{x} \;\le\; \frac{\bar{x}^3}{6} + \bar{x}. \tag{94}$$

Both kernels therefore share one bracket.[1]

**Why the envelope alone does not settle the question.** Nothing beyond $|\rho| \le \rho_{\max}$ enters (90): no monotonicity, no shape, no property of the trajectory but the window length. That generality is also its cost. A constant envelope spreads the forcing over the whole window, whereas both channels are concentrated at its end ($\tau^6$ and $\tau^4$), and the mismatch grows with $x$, i.e. with the *slowest* ramps. Evaluated with $\bar{\rho} = \rho_{\max}$ over the box, (90) returns a relative deviation of 446% at $\dot{M} = 0.10$ N m/s — larger than the signal it bounds. The envelope form is a sanity check on the algebra, not a result.

What restores it is the one structural fact available for free. Both channels rise monotonically from zero at the onset, while the kernel seen as a function of the integration variable, $s \mapsto \dot{h}_\varphi(\tau_{\text{end}} - s)$, *decreases* — the reflection $s \mapsto \tau_{\text{end}} - s$ reversing a cosh that increases in its own argument. Chebyshev's integral inequality for factors of *opposite* monotonicity therefore applies, in the direction needed: $\int_a^b fg \le (b-a)^{-1}\int_a^b f \int_a^b g$, which follows by integrating $\big(f(u) - f(v)\big)\big(g(u) - g(v)\big) \le 0$ over $[a,b]^2$. Applied to (88) it replaces each maximum in (90) by the channel's time average over the window. With the nominal shapes $\rho_\varphi \propto \Lambda(u)^2$ and $\rho_{GE} \propto u\,\Lambda(u)$, $u = C_2\tau$, the amplitudes cancel in the ratio and

$$\begin{aligned} \bar{\rho} &= R_\varphi(x)\,\rho_{\varphi,\max} + R_{GE}(x)\,\rho_{GE,\max}, \\ R_\varphi(x) &= \frac{1}{x}\frac{\int_0^x \Lambda(u)^2 du}{\Lambda(x)^2} \le \tfrac{1}{7}, \qquad R_{GE}(x) = \frac{1}{x}\frac{\int_0^x u\Lambda(u)\,du}{x\,\Lambda(x)} \le \tfrac{1}{5}. \end{aligned} \tag{95}$$

The two constants are simply the orders of the two channels at the onset: a signal going as $\tau^n$ averages to $1/(n{+}1)$ of its endpoint value, and the channels are sixth and fourth order. Neither limit needs an expansion of the kernel. One application of l'Hôpital's rule to $R = \int_0^x f/(xf)$ gives $\lim_{x\to 0} R = \lim f/(f + xf') = 1/\big(1 + n_{\text{eff}}(0^+)\big)$ with $n_{\text{eff}} = uf'/f$ the local order; three further applications, to $(u\cosh u - u)/(\sinh u - u)$, give $\Psi_\varphi(0^+) = 3$ and hence $n_{\text{eff}}(0^+) = 6$ and 4. That these are the suprema follows from rescaling the window, $u = xt$, which turns either ratio into $R(x) = \int_0^1 f(xt)/f(x)\,dt$; the logarithmic derivative of that integrand is $\big[n_{\text{eff}}(xt) - n_{\text{eff}}(x)\big]/x$ with $n_{\text{eff}} = uf'/f$ the local order, so $R$ decreases wherever $n_{\text{eff}}$ rises. Since $n_{\text{eff}} = d\log f/d\log u$, that is the statement that $\log f$ is *convex* in $\log u$, which is stronger than the monotonicity of $f$ and does not follow from it: $1 - e^{-u}$ rises yet has $n_{\text{eff}}$ falling, because a saturating signal averages closer to its endpoint the longer the window. It holds here because both kernels stiffen. Writing $f = \sum a_n u^n$ with $a_n \ge 0$, as both admit, and $S_k = \sum_{n\ge 0} n^k a_n u^n$, so that $uf' = S_1$ and $u^2 f'' = S_2 - S_1$ — all three sums from $n = 0$, since the coefficients $n$ and $n^2 - n$ vanish on exactly the terms differentiating shifts away, and the Cauchy–Schwarz step below needs one common index set —

$$u\,\frac{dn_{\text{eff}}}{du} = \frac{uf'}{f} + \frac{u^2 f''}{f} - \Big(\frac{uf'}{f}\Big)^2 = \frac{S_0 S_2 - S_1^2}{S_0^2} \;\ge\; 0 \tag{95a}$$

by Cauchy–Schwarz on $\{\sqrt{a_n u^n}\}$ and $\{n\sqrt{a_n u^n}\}$, which the sign pattern of the coefficients licenses. The sequences are infinite, so this needs them in $\ell^2$; both kernels are entire, and a power series with infinite radius may be differentiated termwise on the same disc, so $S_1$ and $S_2$ are finite. Two non-zero $a_n$, which both kernels have, make $f > 0$ for $u > 0$ and the inequality strict. That the channels increase is not an added hypothesis but a consequence of the same sign pattern, and is used not here but in the Chebyshev step.

[1] The step matters. $\Lambda(x)$ is known *as an equality* — the tilt cap supplies it — so only the residual linear term needs $\bar{x}$. Substituting $\bar{x}$ into sinh instead is exponentially lossy: at the slowest ramp $\bar{x} = 7.94$ gives $\sinh\bar{x} = 1404$ against $\Lambda + \bar{x} = 91.4$, a factor of fifteen thrown away for nothing.

The same fact signs the derivatives of (95) directly, which is worth recording because doing so from their closed forms would take a twelve-stage differentiation. With $A = \int_0^x f$, so that $R = A/(xf)$, and $xf' = n_{\mathrm{eff}} f$,

$$R'(x) = \frac{xf(x) - \big(1 + n_{\mathrm{eff}}(x)\big)A(x)}{x^2 f(x)}, \tag{95b}$$

and $n_{\mathrm{eff}}$ increasing gives $f(u) \geq f(x)(u/x)^{n_{\mathrm{eff}}(x)}$ on $[0, x]$, whose integral is $A \geq xf/(1 + n_{\mathrm{eff}})$: $f$ is at least as steep as the power law of its local order at $x$, so its average is at least the $1/(n{+}1)$ that power law would give, which is exactly the value $R$ would need for $R'$ to vanish. Hence $R' < 0$ on both channels, and substituting $n_{\mathrm{eff}} = 2x\Lambda'/\Lambda$ and $(\Lambda + x\Lambda')/\Lambda$ into (95b) returns $dR_\varphi/dx = [x\Lambda^3 - (\Lambda + 2x\Lambda')\int_0^x \Lambda^2]/(x^2\Lambda^3)$ and $dR_{GE}/dx = [x^2\Lambda^2 - (2\Lambda + x\Lambda')\int_0^x u\Lambda]/(x^3\Lambda^2)$. Physically, $\Lambda$ stiffens from cubic to exponential and concentrates each channel ever more sharply at the end of the window. Over the flown $x \in [1.79, 5.20]$ the factors are 0.089–0.133 and 0.145–0.191, so (95) gives away 10–60%; the same evaluation that returned 446% returns 41.8% with it in place.

**Relative form: the result, free of $x$.** Dividing the rate bound of (90) by the nominal rate the estimator fits, $\omega_{\mathrm{nom,end}} = C_1(\cosh x - 1)$ with $C_1 = \dot{M}/Wz_{CoM}$, and writing $\Delta M_{\mathrm{win}} = \dot{M}\tau_{\mathrm{end}} = \dot{M}x/C_2$ for the moment increment over the window,

$$\frac{|e_\omega|_{\mathrm{end}}}{\omega_{\mathrm{nom,end}}} \leq \frac{\Psi(x)\,\bar{\rho}}{\Delta M_{\mathrm{win}}}, \qquad \Psi(x) = \frac{x \sinh x}{\cosh x - 1} = x \coth\frac{x}{2}, \tag{96}$$

the closed form following from $\cosh x - 1 = 2\sinh^2(x/2)$ and $\sinh x = 2\sinh(x/2)\cosh(x/2)$; $\Psi \geq 2$ everywhere, which is the inequality the exponent-perturbation argument of Sec. VI-D already uses. Here the two approximations work against each other: $\Psi$ grows with $x$ while the averages of (95) shrink with it, and each product is bounded by its $x \to \infty$ limit, $\frac{1}{2}$ and 1. Both products take the form $\coth(x/2)\int_0^x f/f(x)$, with $f = \Lambda^2$ and $f = u\Lambda$ respectively, so l'Hôpital's rule returns the limit as the reciprocal of $\lim f'/f$, namely $2\Lambda'/\Lambda \to 2$ and $1/u + \Lambda'/\Lambda \to 1$: each constant is the reciprocal of its kernel's exponential growth rate, and the gravity channel gets the smaller one because its kernel is the square of the other's. Both bounds are exact rather than sampled. Substituting (95) and $\Psi = x\coth(x/2)$, the two claims read

$$\int_0^x \Lambda(u)^2 du \leq \tfrac{1}{2}\tanh\tfrac{x}{2}\Lambda(x)^2, \qquad \int_0^x u\,\Lambda(u)\,du \leq x\tanh\tfrac{x}{2}\,\Lambda(x), \tag{96a}$$

with $\Lambda(u) = \sinh u - u$; the half-angle tangent enters because $\tanh(x/2) = \Lambda'(x)/\sinh x$. Each step from (95)–(96) to (96a) divides by a quantity that is positive for $x > 0$ — $\coth(x/2)$, $\Lambda$, $\Lambda^2$ and $\cosh^2(x/2)$ — so (96a) is equivalent to the two claims and not merely sufficient for them.

Both are proved from one corollary of the mean value theorem, used three times and nowhere supplemented. *If $f$ is continuous on $[0, b]$, differentiable on $(0, b)$ with $f' \geq 0$, and $f(0) = 0$, then $f \geq 0$ on $[0, b]$*: for $t \in (0, b]$ the theorem gives $f(t) = f(t) - f(0) = f'(\xi)\,t \geq 0$ for some $\xi \in (0, t)$. Write the first claim of (96a) as $D(x) \triangleq \frac{1}{2}\tanh(x/2)\Lambda(x)^2 - \int_0^x \Lambda^2$, so $D(0) = 0$; differentiating and factoring out $\Lambda > 0$ leaves $D' = \Lambda B$ with $B = \phi/\big(4\cosh^2(x/2)\big)$ and

$$\phi(x) \triangleq x + 2x\cosh x - 3\sinh x, \qquad \phi''(x) = \sinh x + 2x\cosh x \geq 0. \tag{96b}$$

Applying the corollary to $\phi'$ (whose derivative is $\phi'' \geq 0$ and which vanishes at 0) gives $\phi' \geq 0$; applying it to $\phi$ gives $\phi \geq 0$, hence $B \geq 0$ and $D' = \Lambda B \geq 0$; applying it to $D$ gives $D \geq 0$, which is the claim. The second inequality of (96a) needs the corollary only once: its difference $E = x\tanh(x/2)\Lambda - \int_0^x u\Lambda$ has $E(0) = 0$ and, on putting the derivative over $\cosh x + 1$ with $\mathrm{sech}^2(x/2) = 2/(\cosh x + 1)$,

$$E'(x) = \frac{\sinh^2 x - 2x\sinh x + x^2\cosh x}{\cosh x + 1} = \frac{\Lambda(x)^2 + x^2\Lambda'(x)}{\cosh x + 1} \;\geq\; 0, \tag{96c}$$

a sum of two non-negative terms over a positive one, so $E \geq 0$. No monotonicity of the products is used, and the limits $\frac{1}{2}$ and 1 enter the proof nowhere: they are values the constants are held to, and the $x \to \infty$ limit shows only that neither can be lowered. Hence

$$\boxed{\frac{|e_\omega|_{\mathrm{end}}}{\omega_{\mathrm{nom,end}}} \;\leq\; \frac{\frac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max}}{\Delta M_{\mathrm{win}}}, \qquad |\Delta M_{\mathrm{crit}}| \;\leq\; \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}} \tag{97}$$

independently of $x$, $C_2$, $J_P$ and $Wz_{CoM}$: the window length that (92) went to some trouble to bound does not survive into the conclusion, and is needed only to bound $\rho_{GE,\max}$ through (93).

The second inequality of (97) is the same argument carried one step further, to the quantity actually reported: the onset minimises the two-segment residual, so a perturbation of the data displaces the minimiser, and that displacement is what moves $M_{\mathrm{crit}}$. The reduction is written out at (98)–(108), which also accounts for the pre-onset baseline the two-segment cost carries.

**How the pivot inertia enters, and how little of it is needed.** The two bounds want $J_P$ from opposite sides, and the parallel-axis identity (82) supplies both. For $\bar{x}$ an inertia *lower* bound is wanted, which is exactly what (82) gives by discarding $J_{CoM} \geq 0$ — and there the mass cancels outright,

$$C_2^2 = \frac{Wz_{CoM}}{J_P} \leq \frac{W z_{CoM}}{m\big(z_{CoM}^2 + a^2\big)} = \frac{g\, z_{CoM}}{z_{CoM}^2 + a^2}, \qquad a \triangleq l_p + p_{\mathrm{off}},$$

so the ceiling on $C_2$ is pure geometry, with no weight and no inertia in it. It is largest at the low end of $z_{CoM} \in [0.2, 0.3]$ m (the unconstrained maximum $\sqrt{g/2a}$ sits at $z_{CoM} = a$, below that interval), giving $C_2 \leq 5.47\ \mathrm{s}^{-1}$ for roll and 5.87 for pitch, hence $Wz_{CoM}C_2 \leq 47.8$ and 49.7 N m/s at the top of the $z_{CoM}$ range.

For $\Delta M_{\mathrm{win}}$ an inertia *upper* bound is wanted, which (82) as written does not give: $\tau_{\mathrm{end}}$ of (93) grows with $J_P$. Retaining the term (82) discards closes it, $J_P \leq J_{CoM}^{\mathrm{CAD}} + m(z_{CoM,\max}^2 + a^2)$, i.e. 0.426 and 0.398 kg m$^2$. A second and entirely inertia-free ceiling is available as a cross-check, since the excitation window is truncated at the moment cap: $\Delta M_{\mathrm{win}} \leq M_{\mathrm{cap}} - |M_{\mathrm{crit}}|_{\min}$, which is 1.67 and 2.34 N m. The first is the tighter and is the one used.

Two further points fix the evaluation. First, the admissible weight is a range, not a number: $W \in [30.08, 31.59]$ N from unloaded to fully ballasted, and each quantity is read at whichever end is unfavourable for it — $W_{\max}$ for $Wz_{CoM}$, $\rho_{\varphi,\max}$ and $J_P$, $W_{\min}$ for $\omega_{\mathrm{nom}}$ and for converting a threshold error into an offset error. Stating the box at the unloaded weight alone would leave the ballasted configurations outside it, since $Wz_{CoM}$ then reaches 9.48 rather than 9.02 N m. Second, at the tilt cap $x$ is *determined* by the geometry and the ramp rate through $\Lambda(x) = \varphi_{\max} W z_{CoM} C_2 / \dot{M}$, and so is $\Delta M_{\mathrm{win}} = \dot{M}x/C_2$; (92)–(93) exist only to avoid solving that transcendental equation, and must not be put into the denominator of the relative form, which would flatter it. The table below solves for $x$ once and uses the window it implies, quoting $\bar{x}$ alongside to show what the shortcut costs.

With $\varphi_{\max} = 10°$, $\beta_M = 0.0345\ \mathrm{rad}^{-1}$ and $\dot{M} \in [0.10, 1.20]$ N m/s, the whole of (90)–(97) then closes on the box of Sec. VI-D:

| | roll ($M_x$) | | | pitch ($M_y$) | | |
|---|---|---|---|---|---|---|
| $\dot{M}$ [N m/s] | 0.10 | 0.45 | 1.20 | 0.10 | 0.45 | 1.20 |
| $x$ at the cap | 5.18 | 3.80 | 2.99 | 5.21 | 3.83 | 3.02 |
| $\bar{x}$ by (92) | 7.94 | 4.81 | 3.47 | 8.04 | 4.87 | 3.51 |
| $\Delta M_{\mathrm{win}}$ [N m] | 0.103 | 0.339 | 0.712 | 0.099 | 0.329 | 0.692 |
| $\rho_{\varphi,\max}$ by (109) [mN m] | | 68.40 | | | 54.01 | |
| $\rho_{GE,\max}$ [mN m] | 0.62 | 2.04 | 4.29 | 0.60 | 1.98 | 4.16 |
| $\bar{\rho}$ by (95) [mN m] | 6.99 | 8.64 | 9.92 | 5.66 | 7.04 | 8.15 |
| envelope only, by (90) [%] | 396 | 93 | 38 | 335 | 79 | 32 |
| $\lvert e_\omega\rvert/\omega_{\mathrm{nom}}$ by (96) [%] | 35.7 | 10.1 | 4.6 | 30.0 | 8.6 | 3.9 |
| $\lvert e_\omega\rvert$ [deg/s] | 19.8 | 6.2 | 3.2 | 17.3 | 5.4 | 2.8 |
| $\lvert\Delta M_{\mathrm{crit}}\rvert$ by (97) [mN m] | 10.3 | 10.6 | 11.0 | 8.2 | 8.5 | 9.0 |
| $\lvert\Delta\lambda_{\mathrm{off}}\rvert$ [mm] | 0.34 | 0.35 | 0.37 | 0.27 | 0.28 | 0.30 |

The envelope row is the one to read first: at the slowest ramp it exceeds the signal it bounds by four-fold, and only past 0.45 N m/s does it become informative at all. The Chebyshev row below it is a factor of 8 to 11 smaller, and the threshold row is nearly flat in $\dot{M}$ as the $x$-free form requires, only the weak $\rho_{GE}$ dependence surviving. Its worst value over the box is 11.0 mN m for roll and 9.0 for pitch, i.e. 0.37 and 0.30 mm of identified offset, of which the gravity channel supplies 92%. Here $\rho_{\varphi,\max}$ is the exact second-order remainder of the gravity moment rather than its Taylor bound,

$$\rho_\varphi = G(\varphi) - G(0) - G'(0)\varphi = W\big[a(1-\cos\varphi) + z_{CoM}(\sin\varphi - \varphi)\big] \leq \tfrac{1}{2} W a\, \varphi_{\max}^2, \tag{109}$$

the second term being negative; the customary bound on the right is loose by 8–14% over the box and there is no reason to pay it.

**The channel the reviewer asks about is the one best absorbed.** One asymmetry between the two channels is worth recording, because it runs opposite to the concern that motivates them. The estimator span contains $\delta\varphi$, whose nominal shape is $\Lambda$, and the bilinear ground-effect channel goes as $s\Lambda(s)$ — close to a multiple of a span member, so a single scaling already accounts for all but 10% of it and the full span for all but 4%. The gravity remainder goes as $\Lambda(s)^2$, whose exponent is twice as large, and the span cannot follow it: 18% survives. The ground-effect contribution to (97) is thus small twice over — once because $\rho_{GE,\max}$ is itself an order below $\rho_{\varphi,\max}$, and again because what remains of it after the fit re-tunes its own coefficients is smaller still.

Two remarks on robustness. First, $\bar{x}$ depends on the geometry only as $(Wz_{CoM}C_2/\dot{M})^{1/3}$, so a factor-of-two error in $Wz_{CoM}$ moves it by 26% — the chain is least sensitive to exactly the quantity that is least certain. Second, the $C_2$ ceiling above is tight enough to be worth checking against the fits, and one of the ten, 5.667 $\mathrm{s}^{-1}$ for `case_05`/$M_x$, exceeds the roll ceiling 5.468 by 3.6%. The margin is set by $p_{\text{off}}$: taking the arm of (82) as $l_p$ alone raises the ceiling to 5.737 and restores the inequality, which is also the physically correct reading, since $p_{\text{off}}$ bounds the offset rather than displacing the contact point. The ceiling above is evaluated with $l_p + p_{\text{off}}$ only because doing so lowers it and is therefore conservative for $\bar{x}$.

**What the bound covers, and where the realised budget is.** Every quantity above is available before a single run: the weight from a scale, $z_{CoM}$ and the arm from the box of Sec. VI-D, $J_{CoM}$ from CAD, the tilt cap and the ramp rates from the excitation protocol, $\beta_M$ from Sec. VI-B. Nothing measured during the campaign enters it. That is what makes it a pre-flight guarantee rather than a description of the runs that happened to be flown, and it is why a reader applying the method to a different airframe can evaluate it for their own vehicle from the six lines above.

Its scope is worth stating as narrowly as it holds. The bound covers the displacement of the identified onset caused by the two forcing terms the estimator cannot absorb, the gravity remainder and the bilinear ground-effect coupling. It is not a total error budget: ground-contact geometry, the ramp-tracking residual and sensor faults lie outside it. Those, and the same bound re-evaluated at the excursions the runs actually reached, are reported with the realised budget in Sec. VIII-E.

**The bound read as a design rule.** Because $\rho_{\varphi,\max} = \frac{1}{2}Wa\varphi_{\max}^2$ carries about 91% of the total, the tilt cap enters squared, while the ramp rate reaches the result only through $\rho_{GE,\max} \propto \dot{M}^{2/3}$:

| tilt cap | | pivot arm | | ramp rate | |
|---|---|---|---|---|---|
| $\varphi_{\max}$ [deg] | bound [mm] | $a$ [m] | bound [mm] | $\dot{M}$ [N m/s] | bound [mm] |
| 10 | 0.400 | 0.160 | 0.400 | 0.10 | 0.372 |
| 7 | 0.201 | 0.130 | 0.331 | 0.45 | 0.384 |
| 5 | 0.105 | 0.100 | 0.262 | 1.20 | 0.400 |
| 3 | 0.040 | | | 3.00 | 0.429 |

Precision is therefore bought by lowering the tilt cap, not by slowing the ramp: a twelve-fold reduction in $\dot{M}$ moves the bound by 7%, whereas halving $\varphi_{\max}$ quarters it. The 10° cap used here was chosen for onset visibility on the lightly loaded configurations, and the table is what that choice costs.

# VI-B: ramp input tracking (rate limits)

The rotor response identified in Sec. V (Table 4: $\omega_{n,\text{rot}} = 21.74$ rad/s, $\zeta = 0.7814$, $\tau_d = 15.67$ ms) tracks a ramp with the steady-state lag

$$t_{\text{lag}} = \frac{2\zeta}{\omega_{n,\text{rot}}} + \tau_d = 71.9 + 15.7 = 87.6 \text{ ms},$$

so the realised moment trails the command by $\dot{M}\,t_{\text{lag}}$, i.e. 8.8 mN m at 0.10 N m/s and 105 mN m at 1.20 N m/s.

This lag does *not* bias $M_{\text{crit}}$. The identification consumes the *realised* moment, reconstructed from the measured rotor speeds, not the commanded one; a common lag therefore shifts the moment trace and the rate trace together and leaves the moment read at the detected onset unchanged. What the lag does consume is window length, which the sample-count gate enforces.

Two residual effects remain and are both gated. First, the realised *slope* may differ from the commanded rate; the run-level gate rejects $|\dot{M}_{\mathrm{meas}} - \dot{M}_{\mathrm{cmd}}|/\dot{M}_{\mathrm{cmd}} > 3\%$, and since $C_1 = K\dot{M}$, a 3% rate error perturbs the amplitude constant by 3% — well inside the ±20% mis-specification the sensitivity analysis shows to be harmless. Second, the start of the ramp is not linear while the rotor transient decays; with $4/(\zeta\omega_{n,\mathrm{rot}}) =$ 235 ms to 2%, that transient is confined to the low-moment part of the window, which the slope gate excludes by scoring only the segment above the per-axis moment floor. The linearity gate (30 mN m RMSE about the best-fit line) then acts as a fault detector for ramps that are stepped or aborted outright.